Annamaria Letizia Palmiero

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Education

Virginia Polytechnic and State University *Expected May 2031*
Ph.D. Statistics.

University of Maryland, Baltimore County (UMBC) *Expected May 2026*
B.S. Statistics; B.S. Computer Science. Minor: Mathematics. *CGPA: 3.812 / 4.0*

Experience

Research Assistant, University of Maryland, Baltimore County *Jan 2026–Present*

- Conduct statistical research on oyster disease dynamics under the mentorship of Dr. Allison Tracy (Institute of Marine and Environmental Technology & UMBC) and Dr. Yehenew Kifle (UMBC).
- Selected as an inaugural Center for Integrated Mathematical and Biological Sciences Research and Education (CIMBRE) Student Awardee for research at the interface of mathematical and biological sciences.

Research Assistant, Yale University *Jun 2025–Present*

- Conducted mathematical research in computational fluid dynamics focusing on matrix decomposition, dimensionality reduction, and machine-learning prediction of high-speed turbulent flows.
- Co-authoring manuscript on decomposition techniques for characterizing vortex structures.

Student Intern, Department of the U.S. Navy *Jan 2024–Present*

- Developed an ordinal logistic regression model using quarterly data to model project heath indicators for senior Command leadership.
- Validated and documented enterprise data infrastructure by auditing a Navy-wide Data Dictionary, assessed data completeness and cross-platform accessibility, and authored documentation tracking field-level mappings to secondary legacy systems to establish data lineage.
- Designed and deployed a Power Apps–SharePoint system with automated Power Automate workflows for the Department Administrative Officer as the sole developer.

Research Assistant, UMBC Division of Information Technology *Jan–May 2024*

- Maintained predictive and inferential models (machine learning, logistic/linear regression) for student success analytics and enrollment forecasting.
- Performed statistical analyses, validated data pipelines, and visualized trends to inform decision-making and optimize predictive models.

Self-Directed Researcher, UMBC *Jan–May 2023*

- Developed supplemental programming challenges and quantitatively analyzed their effect on student performance in an introductory computing course.
- Conducted statistical analysis in R, and presented findings at UMBC's Teaching & Learning Symposium.

Publications

Mai Al Shaaban, Joey Takei, **Annamaria Palmiero**, Leya Dereje. "Investigation of Decomposition Techniques for Characterizing Complex Vortex Structures in MVG Controlled Boundary Layer." – Under Review by *Computation* (MDPI).

Conference Presentations

Oral Presentations

- **Matrix Decomposition for Characterizing Vortex Structures**
  Young Mathematicians Conference, Ohio State University (Aug 2025).
- **Matrix Decomposition in Computational Fluid Dynamics**
  Yale University–Williams College Undergraduate Mathematics Research Conference (Jul 2025).
- **Exploring Research with R: From Data to Insight**
  STEM Student Leadership Conference, Loyola University (May 2025).
- **Visualizations in Processing**
  UMBC Programming Lecture Series (Apr 2025).

Poster Presentations

- **Statistical Modeling of Parasite Infection Risk in *Crassostrea virginica***
  Undergraduate Research and Creative Achievement Day, UMBC (April 2026).
- **Statistical Modeling of Parasite Infection Risk in *Crassostrea virginica***
  First Annual Center for Integrated Mathematical and Biological sciences Research and Education Symposium, UMBC (April 2026).
- **Matrix Decomposition for Complex Vortex Structure Characterization**
  Joint Mathematics Meeting, Washington D.C. (Jan 2026).
- **Machine Learning–Based Prediction for Vortex Structures in High–Speed Flows**
  MIT Undergraduate Research Technology Conference (Oct 2025).
- **Matrix Decomposition & Prediction for Vortex Structure Characterization**
  Summer Undergraduate Research Festival, UMBC (Aug 2025).
- **Machine Learning–Based Vortex Evolution**
  Jane Street Summer Mathematics Day (Jul 2025).
- **The Effects of External Learning on Computer Science Performance**
  UMBC Teaching & Learning Symposium (Apr 2023).

Grants, Fellowships & Honors

- **CIMBRE Undergraduate Research Award**, $2,000
- **National Science Foundation Travel Grant**, $500 — Young Mathematicians Conference
- **Whiting-Turner Research Grant**, $2,000
- **UMBC Mathematics and Statistics Department Research Grant**, $2,000
- **France-Merrick Fellowship**, $8,500 – Commitment to Service and Leadership (2023–2024)
- **Phi Beta Kappa** (Inducted 2025); **Pi Mu Epsilon** (2025); **Mu Sigma Rho** (2024)
- **Center for the Integration of Research, Teaching and Learning (CIRTL) Associate** (Inducted 2023)
- **College of Natural and Mathematical Sciences Scholar** (2023–Present)
- **President's List** (Fall 2024, Spring 2025, Fall 2025); **Dean's List** (Fall 2023)

Teaching Experience

- **Lead Teaching Fellow**, Computational Thinking and Design — Aug 2024–Present
- **Teaching Assistant**, Multivariable Calculus — Aug 2023–Dec 2023
- **Teaching Fellow**, Computational Thinking and Design — Aug 2022–May 2023

Leadership

- **President**, Mu Sigma Rho Statistics Honors Society (2025–Present)
- **Vice President**, Mu Sigma Rho (2024–2025)
- **Service Site Leader**, Baltimore Animal Rescue and Care Shelter (2024–2025)
- **Treasurer & Volunteer Coordinator**, Animal Welfare Student Organization (2023–2025)
- **Living Learning Community Peer Leader & Mentor** (2022–2025)

Skills

- **Statistical:** Probability theory, statistical inference, regression theory, multivariate statistics, statistical computing, stochastic processes, data cleaning, visualization.
- **Computational:** Machine learning, algorithm analysis, data structures, automata theory, efficient coding practices, object-oriented design.
- **Programming:** R, Python, C++, C, LaTeX, Assembly, MATLAB, Processing.
- **Software:** Power Apps, Power Automate, Power BI, Tableau, Excel, Word, PowerPoint.